

Лабораторная работа №4

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

2026-02-16

1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Приобретение практических навыков установки и базового конфигурирования HTTP-сервера Apache

Раздел 2

2. Выполнение лабораторной работы

2. Выполнение лабораторной работы

2.0.1 Установка HTTP-сервера

- Выполнен `yum grouplist`

```
[root@server.bahis.net server]# LANG=C yum grouplist
Rocky Linux 9 - BaseOS                8.9 kB/s | 4.3 kB      00:00
Rocky Linux 9 - AppStream              15 kB/s | 4.8 kB      00:00
Rocky Linux 9 - Extras                 9.7 kB/s | 3.1 kB      00:00
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  KDE Plasma Workspaces
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Fedora Packager
```

2. Выполнение лабораторной работы

2.0.1 Установка HTTP-сервера

- Выполнен `yum grouplist`
- Обнаружена группа Basic Web Server

```
[root@server.bahis.net server]# LANG=C yum grouplist
Rocky Linux 9 - BaseOS                8.9 kB/s | 4.3 kB      00:00
Rocky Linux 9 - AppStream             15 kB/s | 4.8 kB      00:00
Rocky Linux 9 - Extras                9.7 kB/s | 3.1 kB      00:00
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  KDE Plasma Workspaces
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Fedora Packager
```

2. Выполнение лабораторной работы

2.0.1 Установка HTTP-сервера

- Выполнен `yum grouplist`
- Обнаружена группа `Basic Web Server`
- Установлена среда `Server with GUI`

```
[root@server.bahis.net server]# LANG=C yum grouplist
Rocky Linux 9 - BaseOS                8.9 kB/s | 4.3 kB      00:00
Rocky Linux 9 - AppStream              15 kB/s | 4.8 kB      00:00
Rocky Linux 9 - Extras                 9.7 kB/s | 3.1 kB      00:00
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  KDE Plasma Workspaces
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Fedora Packager
```

- Выполнена установка группы Basic Web Server

```
[root@server.bahis.net server]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:00:14 ago on Tue 10 Feb 2026 02:21:33 PM UTC.
Dependencies resolved.
=====
Package                                Architecture Version                                Repository                               Size
=====
Installing group/module packages:
httpd                                  x86_64      2.4.62-7.el9_7.3                        appstream                                45 k
httpd-manual                           noarch      2.4.62-7.el9_7.3                        appstream                                2.2 M
mod_fcgid                               x86_64      2.3.9-28.el9                            appstream                                74 k
mod_ssl                                x86_64      1:2.4.62-7.el9_7.3                      appstream                                110 k
Installing dependencies:
apr                                     x86_64      1.7.0-12.el9_3                          appstream                                122 k
apr-util                               x86_64      1.6.1-23.el9                            appstream                                94 k
apr-util-bdb                           x86_64      1.6.1-23.el9                            appstream                                12 k
httpd-core                              x86_64      2.4.62-7.el9_7.3                        appstream                                1.4 M
httpd-filesystem                        noarch      2.4.62-7.el9_7.3                        appstream                                12 k
httpd-tools                             x86_64      2.4.62-7.el9_7.3                        appstream                                79 k
rocky-logos-httpd                       noarch      90.16-1.el9                             appstream                                24 k
Installing weak dependencies:
apr-util-openssl                        x86_64      1.6.1-23.el9                            appstream                                14 k
mod_http2                               x86_64      2.0.26-5.el9                            appstream                                163 k
mod_lua                                 x86_64      2.4.62-7.el9_7.3                        appstream                                59 k
Installing Groups:
Basic Web Server

Transaction Summary
=====
```

- Выполнена установка группы Basic Web Server
- Установлены пакеты httpd, mod_ssl, mod_fcgid

```
[root@server.bahis.net server]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:00:14 ago on Tue 10 Feb 2026 02:21:33 PM UTC.
Dependencies resolved.
=====
Package                Architecture Version                      Repository                Size
=====
Installing group/module packages:
httpd                  x86_64      2.4.62-7.el9_7.3            appstream                 45 k
httpd-manual           noarch      2.4.62-7.el9_7.3            appstream                 2.2 M
mod_fcgid              x86_64      2.3.9-28.el9                appstream                 74 k
mod_ssl               x86_64      1:2.4.62-7.el9_7.3          appstream                 110 k
Installing dependencies:
apr                   x86_64      1.7.0-12.el9_3              appstream                 122 k
apr-util             x86_64      1.6.1-23.el9                appstream                 94 k
apr-util-bdb         x86_64      1.6.1-23.el9                appstream                 12 k
httpd-core           x86_64      2.4.62-7.el9_7.3            appstream                 1.4 M
httpd-filesystem     noarch      2.4.62-7.el9_7.3            appstream                 12 k
httpd-tools          x86_64      2.4.62-7.el9_7.3            appstream                 79 k
rocky-logos-httpd    noarch      90.16-1.el9                  appstream                 24 k
Installing weak dependencies:
apr-util-openssl     x86_64      1.6.1-23.el9                appstream                 14 k
mod_http2            x86_64      2.0.26-5.el9                appstream                 163 k
mod_lua              x86_64      2.4.62-7.el9_7.3            appstream                 59 k
Installing Groups:
Basic Web Server

Transaction Summary
=====
```

- Выполнена установка группы Basic Web Server
- Установлены пакеты httpd, mod_ssl, mod_fcgid
- Установлены зависимости apr, apr-util

```
[root@server.bahis.net server]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:00:14 ago on Tue 10 Feb 2026 02:21:33 PM UTC.
Dependencies resolved.
=====
Package                Architecture Version                Repository            Size
=====
Installing group/module packages:
httpd                  x86_64        2.4.62-7.el9_7.3      appstream             45 k
httpd-manual           noarch        2.4.62-7.el9_7.3      appstream             2.2 M
mod_fcgid              x86_64        2.3.9-28.el9          appstream             74 k
mod_ssl               x86_64        1:2.4.62-7.el9_7.3    appstream            110 k
Installing dependencies:
apr                   x86_64        1.7.0-12.el9_3        appstream            122 k
apr-util             x86_64        1.6.1-23.el9          appstream            94 k
apr-util-bdb         x86_64        1.6.1-23.el9          appstream            12 k
httpd-core            x86_64        2.4.62-7.el9_7.3      appstream            1.4 M
httpd-filesystem      noarch        2.4.62-7.el9_7.3      appstream            12 k
httpd-tools           x86_64        2.4.62-7.el9_7.3      appstream            79 k
rocky-logos-httpd     noarch        90.16-1.el9           appstream            24 k
Installing weak dependencies:
apr-util-openssl      x86_64        1.6.1-23.el9          appstream            14 k
mod_http2             x86_64        2.0.26-5.el9          appstream            163 k
mod_lua               x86_64        2.4.62-7.el9_7.3      appstream            59 k
Installing Groups:
Basic Web Server

Transaction Summary
=====
```

2.1 Базовое конфигурирование HTTP-сервера

- Просмотрен каталог `/etc/httpd/conf.d`



```
[root@server.bahis.net server]# ls /etc/httpd/conf
httpd.conf  magic
```

Рисунок 3: Содержимое каталога `/etc/httpd/conf.d`

2.1 Базовое конфигурирование HTTP-сервера

- Просмотрен каталог `/etc/httpd/conf.d`
- Обнаружены `ssl.conf`, `welcome.conf`, `userdir.conf`



```
[root@server.bahis.net server]# ls /etc/httpd/conf
httpd.conf  magic
```

Рисунок 3: Содержимое каталога `/etc/httpd/conf.d`

2.1 Базовое конфигурирование HTTP-сервера

- Просмотрен каталог `/etc/httpd/conf.d`
- Обнаружены `ssl.conf`, `welcome.conf`, `userdir.conf`
- Подтверждена стандартная конфигурация Apache



```
[root@server.bahis.net server]# ls /etc/httpd/conf
httpd.conf  magic
```

Рисунок 3: Содержимое каталога `/etc/httpd/conf.d`

- В firewalld добавлена служба http

```
[root@server.bahis.net server]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns ssh
[root@server.bahis.net server]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcup
sd audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage
e bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph
-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb
dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-re
gistry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman
foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp
galera ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 ht
tps ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin
kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-contr
ol-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure
kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-
readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-clie
nt llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt
mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea
-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconso
le plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus promethe
us-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius
rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-c
lient samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing s
yncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc t
or-socks transmission-client upnp-client vds vnc-server warpinator wbem-http wbem-https
wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans
xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server.bahis.net server]# firewall-cmd --add-service=http
success
[root@server.bahis.net server]# firewall-cmd --add-service=http --permanent
```

- В firewalld добавлена служба http
- Правило добавлено временно и постоянно

```
[root@server.bahis.net server]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns ssh
[root@server.bahis.net server]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcup
sd audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage
e bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph
-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb
dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-re
gistry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman
foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp
galera ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 ht
tps ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin
kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-contr
ol-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure
kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet
-readyonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-clie
nt llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt
mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea
-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconso
le plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus promethe
us-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius
rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-c
lient samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing s
yncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc t
or-socks transmission-client upnp-client vds vnc-server warpinator wbem-http wbem-https
wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans
xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server.bahis.net server]# firewall-cmd --add-service=http
success
[root@server.bahis.net server]# firewall-cmd --add-service=http --permanent
```

- Запущен мониторинг `journalctl -x -f`

```
root@server.bahis.net server]# journalctl -x -f
eb 10 14:21:57 server.bahis.net dnf[8486]: Extra Packages for Enterprise Linux 9 - x86_64
6.8 MB/s | 20 MB    00:02
eb 10 14:22:04 server.bahis.net dnf[8486]: Extra Packages for Enterprise Linux 9 openh26
2.0 kB/s | 993 B    00:00
eb 10 14:22:04 server.bahis.net dnf[8486]: Rocky Linux 9 - BaseOS
12 kB/s | 4.3 kB    00:00
eb 10 14:22:05 server.bahis.net dnf[8486]: Rocky Linux 9 - AppStream
15 kB/s | 4.8 kB    00:00
eb 10 14:22:05 server.bahis.net dnf[8486]: Rocky Linux 9 - Extras
9.9 kB/s | 3.1 kB    00:00
eb 10 14:22:06 server.bahis.net dnf[8486]: Metadata cache created.
eb 10 14:22:06 server.bahis.net systemd[1]: dnf-makecache.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit dnf-makecache.service has successfully entered the 'dead' state.
eb 10 14:22:06 server.bahis.net systemd[1]: Finished dnf makecache.
Subject: A start job for unit dnf-makecache.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit dnf-makecache.service has finished successfully.

The job identifier is 2990.
eb 10 14:22:06 server.bahis.net systemd[1]: dnf-makecache.service: Consumed 6.983s CPU time.
Subject: Resources consumed by unit runtime
Defined-By: systemd
```

- Запущен мониторинг `journalctl -x -f`
- Контроль состояния службы в реальном времени

```
root@server.bahis.net server]# journalctl -x -f
eb 10 14:21:57 server.bahis.net dnf[8486]: Extra Packages for Enterprise Linux 9 - x86_64
6.8 MB/s | 20 MB    00:02
eb 10 14:22:04 server.bahis.net dnf[8486]: Extra Packages for Enterprise Linux 9 openh26
2.0 kB/s | 993 B    00:00
eb 10 14:22:04 server.bahis.net dnf[8486]: Rocky Linux 9 - BaseOS
12 kB/s | 4.3 kB    00:00
eb 10 14:22:05 server.bahis.net dnf[8486]: Rocky Linux 9 - AppStream
15 kB/s | 4.8 kB    00:00
eb 10 14:22:05 server.bahis.net dnf[8486]: Rocky Linux 9 - Extras
9.9 kB/s | 3.1 kB    00:00
eb 10 14:22:06 server.bahis.net dnf[8486]: Metadata cache created.
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A start job for unit dnf-makecache.service has finished successfully.

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eb 10 14:22:06 server.bahis.net systemd[1]: dnf-makecache.service: Consumed 6.983s CPU time.
Subject: Resources consumed by unit runtime
Defined-By: systemd
```

- Выполнены `systemctl enable httpd` и `start httpd`

```
[root@server.bahis.net ~]# systemctl enable httpd
[root@server.bahis.net ~]# systemctl start httpd
[root@server.bahis.net ~]# systemctl stutas httpd
Unknown command verb stutas.
[root@server.bahis.net ~]# systemctl status httpd
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-02-10 14:25:30 UTC; 6min ago
     Docs: man:httpd.service(8)
  Main PID: 8894 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/s
    Tasks: 178 (limit: 4498)
    Memory: 22.7M (peak: 23.0M)
      CPU: 164ms
    CGroup: /system.slice/httpd.service
            └─8894 /usr/sbin/httpd -DFOREGROUND
              └─8895 /usr/sbin/httpd -DFOREGROUND
                └─8896 /usr/sbin/httpd -DFOREGROUND
                  └─8897 /usr/sbin/httpd -DFOREGROUND
                    └─8898 /usr/sbin/httpd -DFOREGROUND
                      └─8899 /usr/sbin/httpd -DFOREGROUND

Feb 10 14:25:30 server.bahis.net systemd[1]: Starting The Apache HTTP Server...
Feb 10 14:25:30 server.bahis.net systemd[1]: Started The Apache HTTP Server.
Feb 10 14:25:30 server.bahis.net httpd[8894]: Server configured, listening on: port 443.
```

Рисунок 6: Состояние службы httpd после запуска

- Выполнены `systemctl enable httpd` и `start httpd`
- Статус — `active (running)`

```
[root@server.bahis.net ~]# systemctl enable httpd
[root@server.bahis.net ~]# systemctl start httpd
[root@server.bahis.net ~]# systemctl stutas httpd
Unknown command verb stutas.
[root@server.bahis.net ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-02-10 14:25:30 UTC; 6min ago
     Docs: man:httpd.service(8)
  Main PID: 8894 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/s
    Tasks: 178 (limit: 4498)
   Memory: 22.7M (peak: 23.0M)
      CPU: 164ms
   CGroup: /system.slice/httpd.service
           └─8894 /usr/sbin/httpd -DFOREGROUND
             └─8895 /usr/sbin/httpd -DFOREGROUND
               └─8896 /usr/sbin/httpd -DFOREGROUND
                 └─8897 /usr/sbin/httpd -DFOREGROUND
                   └─8898 /usr/sbin/httpd -DFOREGROUND
                     └─8899 /usr/sbin/httpd -DFOREGROUND

Feb 10 14:25:30 server.bahis.net systemd[1]: Starting The Apache HTTP Server...
Feb 10 14:25:30 server.bahis.net systemd[1]: Started The Apache HTTP Server.
Feb 10 14:25:30 server.bahis.net httpd[8894]: Server configured, listening on: port 443.
```

Рисунок 6: Состояние службы `httpd` после запуска

- Выполнены `systemctl enable httpd` и `start httpd`
- Статус — `active (running)`
- Прослушивание портов 80 и 443

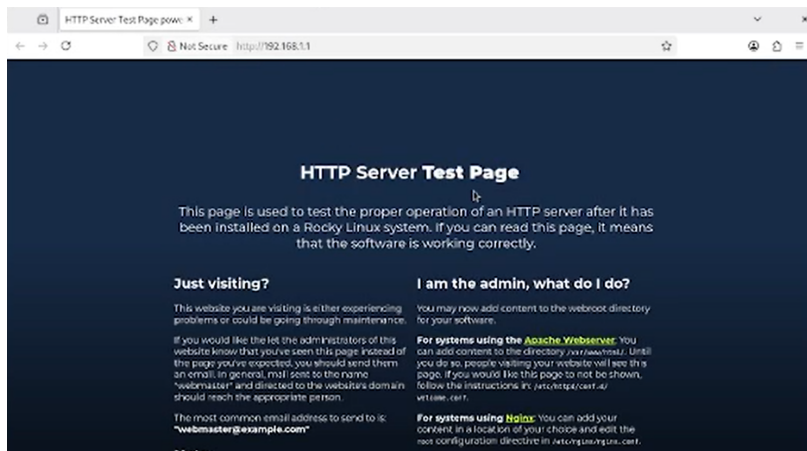
```
[root@server.bahis.net ~]# systemctl enable httpd
[root@server.bahis.net ~]# systemctl start httpd
[root@server.bahis.net ~]# systemctl stutas httpd
Unknown command verb stutas.
[root@server.bahis.net ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-02-10 14:25:30 UTC; 6min ago
     Docs: man:httpd.service(8)
  Main PID: 8894 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/s
    Tasks: 178 (limit: 4498)
   Memory: 22.7M (peak: 23.0M)
      CPU: 164ms
    CGroup: /system.slice/httpd.service
            └─8894 /usr/sbin/httpd -DFOREGROUND
              └─8895 /usr/sbin/httpd -DFOREGROUND
                └─8896 /usr/sbin/httpd -DFOREGROUND
                  └─8897 /usr/sbin/httpd -DFOREGROUND
                    └─8898 /usr/sbin/httpd -DFOREGROUND
                      └─8899 /usr/sbin/httpd -DFOREGROUND

Feb 10 14:25:30 server.bahis.net systemd[1]: Starting The Apache HTTP Server...
Feb 10 14:25:30 server.bahis.net systemd[1]: Started The Apache HTTP Server.
Feb 10 14:25:30 server.bahis.net httpd[8894]: Server configured, listening on: port 443.
```

Рисунок 6: Состояние службы httpd после запуска

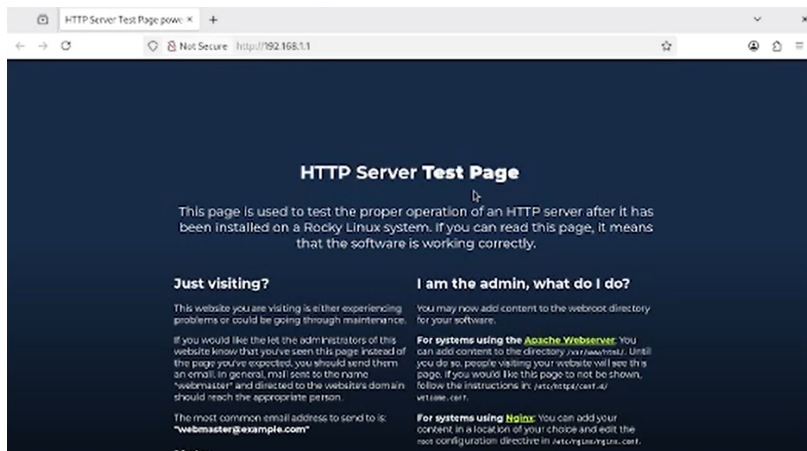
2.2 Анализ работы HTTP-сервера

- На клиенте открыт `http://192.168.1.1`



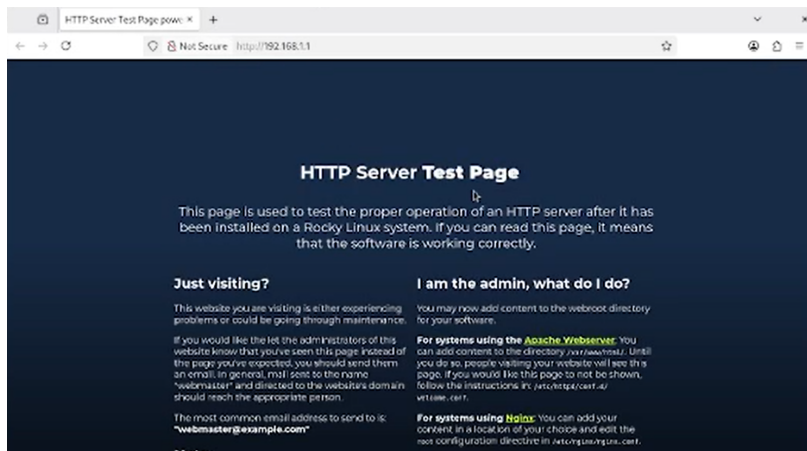
2.2 Анализ работы HTTP-сервера

- На клиенте открыт `http://192.168.1.1`
- Отобрана страница HTTP Server Test Page



2.2 Анализ работы HTTP-сервера

- На клиенте открыт `http://192.168.1.1`
- Отобразилась страница HTTP Server Test Page
- Подтверждена работа Apache



- Выполнен мониторинг /var/log/httpd/access_log

```
[root@server.bahis.net ~]# tail -f /var/log/httpd/access_log
192.168.1.30 - - [10/Feb/2026:18:53:42 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0
(X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15
443 "http://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Fire
fox/140.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "ht
tp://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140
.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http:
//192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
^C
+
[root@server.bahis.net ~]#
```

Рисунок 8: Мониторинг access_log веб-сервера

- Выполнен мониторинг /var/log/httpd/access_log
- Зафиксированы GET-запросы от 192.168.1.30

```
[root@server.bahis.net ~]# tail -f /var/log/httpd/access_log
192.168.1.30 - - [10/Feb/2026:18:53:42 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0
(X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15
443 "http://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Fire
fox/140.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "ht
tp://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140
.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http:
//192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
^C
+
[root@server.bahis.net ~]#
```

Рисунок 8: Мониторинг access_log веб-сервера

- Выполнен мониторинг /var/log/httpd/access_log
- Зафиксированы GET-запросы от 192.168.1.30
- Коды ответа 200 и 404

```
[root@server.bahis.net ~]# tail -f /var/log/httpd/access_log
192.168.1.30 - - [10/Feb/2026:18:53:42 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0
(X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15
443 "http://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Fire
fox/140.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "ht
tp://192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140
.0"
192.168.1.30 - - [10/Feb/2026:18:53:43 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http:
//192.168.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
^C
+
[root@server.bahis.net ~]#
```

Рисунок 8: Мониторинг access_log веб-сервера

- Выполнен мониторинг /var/log/httpd/error_log

```
[root@server.bahis.net ~]# tail -f /var/log/httpd/error_log
[Tue Feb 10 14:25:30.864195 2026] [core:notice] [pid 8894:tid 8894] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0
[Tue Feb 10 14:25:30.866450 2026] [suexec:notice] [pid 8894:tid 8894] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)
[Tue Feb 10 14:25:30.878604 2026] [lbmethod_heartbeat:notice] [pid 8894:tid 8894] AH02282: No slotmem from mod_heartbeat
[Tue Feb 10 14:25:30.884986 2026] [mpm_event:notice] [pid 8894:tid 8894] AH00489: Apache/2.4.62 (Rocky Linux) OpenSSL/3.5.1 mod_fcgid/2.3.9 configured -- resuming normal operations
[Tue Feb 10 14:25:30.884999 2026] [core:notice] [pid 8894:tid 8894] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'
^C
[root@server.bahis.net ~]#
```

Рисунок 9: Мониторинг Error_log веб-сервера

- Выполнен мониторинг /var/log/httpd/error_log
- Сообщение об отсутствии index.html

```
[root@server.bahis.net ~]# tail -f /var/log/httpd/error_log
[Tue Feb 10 14:25:30.864195 2026] [core:notice] [pid 8894:tid 8894] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0
[Tue Feb 10 14:25:30.866450 2026] [suexec:notice] [pid 8894:tid 8894] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)
[Tue Feb 10 14:25:30.878604 2026] [lbmethod_heartbeat:notice] [pid 8894:tid 8894] AH02282: No slotmem from mod_heartbeat
[Tue Feb 10 14:25:30.884986 2026] [mpm_event:notice] [pid 8894:tid 8894] AH00489: Apache/2.4.62 (Rocky Linux) OpenSSL/3.5.1 mod_fcgid/2.3.9 configured -- resuming normal operations
[Tue Feb 10 14:25:30.884999 2026] [core:notice] [pid 8894:tid 8894] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'
^C
[root@server.bahis.net ~]#
```

Рисунок 9: Мониторинг Error_log веб-сервера

- Выполнен мониторинг /var/log/httpd/error_log
- Сообщение об отсутствии index.html
- Код ответа 403

```
[root@server.bahis.net ~]# tail -f /var/log/httpd/error_log
[Tue Feb 10 14:25:30.864195 2026] [core:notice] [pid 8894:tid 8894] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0
[Tue Feb 10 14:25:30.866450 2026] [suexec:notice] [pid 8894:tid 8894] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)
[Tue Feb 10 14:25:30.878604 2026] [lbmethod_heartbeat:notice] [pid 8894:tid 8894] AH02282: No slotmem from mod_heartbeat
[Tue Feb 10 14:25:30.884986 2026] [mpm_event:notice] [pid 8894:tid 8894] AH00489: Apache/2.4.62 (Rocky Linux) OpenSSL/3.5.1 mod_fcgid/2.3.9 configured -- resuming normal operations
[Tue Feb 10 14:25:30.884999 2026] [core:notice] [pid 8894:tid 8894] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'
^C
[root@server.bahis.net ~]#
```

Рисунок 9: Мониторинг Error_log веб-сервера

2.3 Настройка виртуального хостинга для HTTP-сервера

- В прямую зону добавлена запись `www A 192.168.1.1`

```
$ORIGIN.
$TTL 86400      ; 1 day
bahis.net. IN SOA bahis.net.  server.bahis.net. (
                2026020919 ; serial
                86400      ; refresh (1 day)
                3600       ; retry (1 hour)
                604800     ; expire (1 week)
                10800      ; minimum (3 hours)
                )
                NS      server.bahis.net.
                A      192.168.1.1
$ORIGIN bahis.net.
$TTL 2400       ; 40 minutes
client A      192.168.1.30
      DHCID   (AAEBpkVVD5B7g]AKQJJsXpXqCDygKsftukn20854qA0pp
      IIW= ) ; 1 1 32
$TTL 86400     ; 1 day
server A      192.168.1.1
ns A      192.168.1.1
dhcp A      192.168.1.1
www A      192.168.1.1
```

2.3 Настройка виртуального хостинга для HTTP-сервера

- В прямую зону добавлена запись `www A 192.168.1.1`
- Присутствует запись клиента с DHCID

```
$ORIGIN.  
$TTL 86400          ; 1 day  
bahis.net. IN SOA bahis.net. server.bahis.net. (  
                2026020919 ; serial  
                86400      ; refresh (1 day)  
                3600       ; retry (1 hour)  
                604800     ; expire (1 week)  
                10800      ; minimum (3 hours)  
                )  
                NS server.bahis.net.  
                A 192.168.1.1  
$ORIGIN bahis.net.  
$TTL 2400           ; 40 minutes  
client A 192.168.1.30  
      DHCID (AAEBpkVVD5B7g]AKQJsXpXqCDygKsftukn20854qA0pp  
            IIW= ) ; 1 1 32  
$TTL 86400          ; 1 day  
server A 192.168.1.1  
ns A 192.168.1.1  
dhcp A 192.168.1.1  
www A 192.168.1.1
```

- В обратную зону добавлена PTR-запись

```
GNU nano 5.6.1 /var/named/master/rz/192.168.1 Modified
ORIGIN.
$TTL 86400 ; 1 day
1.168.192.in-addr.arpa. IN SOA bahis.net. server.bahis.net. (
    2026020109 ; serial
    86400 ; refresh (1 day)
    3600 ; retry (1 hour)
    604800 ; expire (1 week)
    10800 ; minimum (3 hours)
)
    NS server.bahis.net.
    A 192.168.1.1
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.bahis.net.
PTR ns.bahis.net.
PTR dhcp.bahis.net.
1 PTR www.bahis.net.
$TTL 2400 ; 40 minutes
30 PTR client.bahis.net.
DHCID (AAEBpkVVD5B7g]AKQJsXpXqCDygKsftukn20854qA0pp
IIW= ) ; 1 1 32
```

Рисунок 11: Файл обратной DNS-зоны 192.168.1 с PTR-записью

- В обратную зону добавлена PTR-запись
- 192.168.1.1 → www.bahis.net.

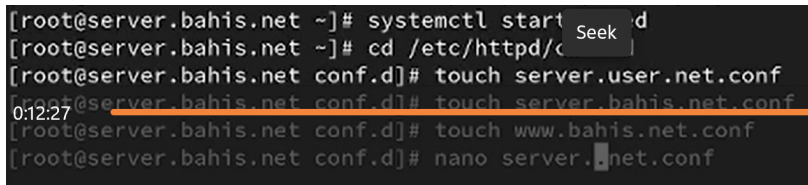
```

GNU nano 5.6.1 /var/named/master/rz/192.168.1 Modified
ORIGIN.
$TTL 86400 ; 1 day
1.168.192.in-addr.arpa. IN SOA bahis.net. server.bahis.net. (
    2026020109 ; serial
    86400 ; refresh (1 day)
    3600 ; retry (1 hour)
    604800 ; expire (1 week)
    10800 ; minimum (3 hours)
)
    NS server.bahis.net.
    A 192.168.1.1
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.bahis.net.
PTR ns.bahis.net.
PTR dhcp.bahis.net.
1 PTR www.bahis.net.
$TTL 2400 ; 40 minutes
30 PTR client.bahis.net.
DHCID (AAEBpkVVD5B7g]AKQJsXpXqCDygKsftukn20854qA0pp
IIW= ) ; 1 1 32

```

Рисунок 11: Файл обратной DNS-зоны 192.168.1 с PTR-записью

- Созданы файлы виртуальных хостов

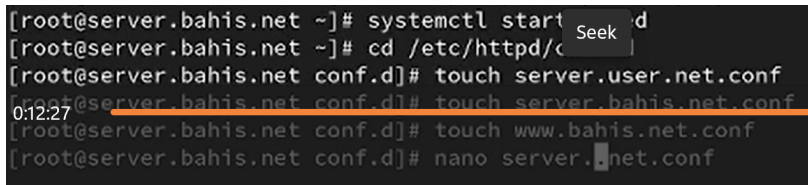


A screenshot of a terminal window showing a series of commands to create virtual host configuration files. The terminal has a dark background with light-colored text. A red horizontal line highlights the command `touch server.bahis.net.conf`. A mouse cursor is visible over the word `server` in the command `touch server.bahis.net.conf`. A timestamp `0:12:27` is visible on the left side of the terminal. A semi-transparent box with the word `Seek` is overlaid on the terminal text.

```
[root@server.bahis.net ~]# systemctl start d
[root@server.bahis.net ~]# cd /etc/httpd/conf.d
[root@server.bahis.net conf.d]# touch server.user.net.conf
0:12:27 [root@server.bahis.net conf.d]# touch server.bahis.net.conf
[root@server.bahis.net conf.d]# touch www.bahis.net.conf
[root@server.bahis.net conf.d]# nano server.bahis.net.conf
```

Рисунок 12: Создание конфигурационных файлов виртуальных хостов

- Созданы файлы виртуальных хостов
- `server.bahis.net.conf`



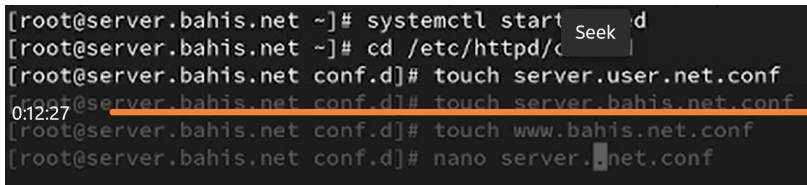
A terminal window screenshot with a dark background and white text. The commands and their outputs are as follows:

```
[root@server.bahis.net ~]# systemctl start d
[root@server.bahis.net ~]# cd /etc/httpd/conf.d
[root@server.bahis.net conf.d]# touch server.user.net.conf
[root@server.bahis.net conf.d]# touch server.bahis.net.conf
[root@server.bahis.net conf.d]# touch www.bahis.net.conf
[root@server.bahis.net conf.d]# nano server.bahis.net.conf
```

A red horizontal line is drawn under the command `touch server.bahis.net.conf`. A white timestamp `0:12:27` is visible on the left side of the terminal window. A semi-transparent grey box with the word "Seek" is positioned over the top right portion of the terminal output.

Рисунок 12: Создание конфигурационных файлов виртуальных хостов

- Созданы файлы виртуальных хостов
- `server.bahis.net.conf`
- `www.bahis.net.conf`



A terminal window screenshot showing the following commands and their outputs:

```
[root@server.bahis.net ~]# systemctl start d
[root@server.bahis.net ~]# cd /etc/httpd/conf.d
[root@server.bahis.net conf.d]# touch server.user.net.conf
[root@server.bahis.net conf.d]# touch server.bahis.net.conf
[root@server.bahis.net conf.d]# touch www.bahis.net.conf
[root@server.bahis.net conf.d]# nano server.bahis.net.conf
```

The line `[root@server.bahis.net conf.d]# touch server.bahis.net.conf` is highlighted with an orange underline. A red horizontal line is positioned below the terminal output. A 'Seek' button is visible in the top right corner of the terminal window. A timestamp '0:12:27' is displayed on the left side of the terminal window.

Рисунок 12: Создание конфигурационных файлов виртуальных хостов

- В server.bahis.net.conf задан <VirtualHost *:80>

```
GNU nano 5.6.1 server.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/server.bahis.net
ServerName server.bahis.net
ErrorLog logs/server.bahis.net-error_log
CustomLog logs/server.bahis.net-access_log common
</VirtualHost>

I
```

Рисунок 13: Конфигурация виртуального хоста server.bahis.net

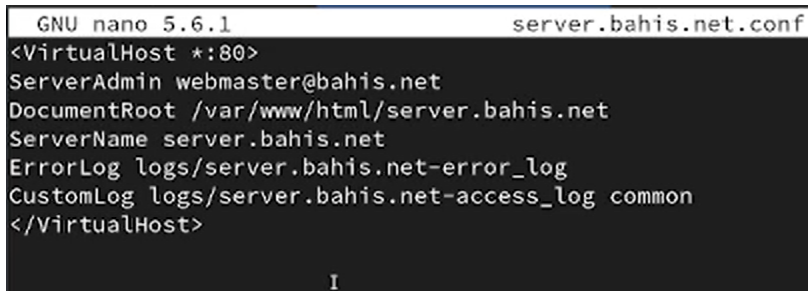
- В server.bahis.net.conf задан <VirtualHost *:80>
- Указаны ServerName, DocumentRoot

```
GNU nano 5.6.1 server.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/server.bahis.net
ServerName server.bahis.net
ErrorLog logs/server.bahis.net-error_log
CustomLog logs/server.bahis.net-access_log common
</VirtualHost>

I
```

Рисунок 13: Конфигурация виртуального хоста server.bahis.net

- В server.bahis.net.conf задан <VirtualHost *:80>
- Указаны ServerName, DocumentRoot
- Настроены отдельные лог-файлы

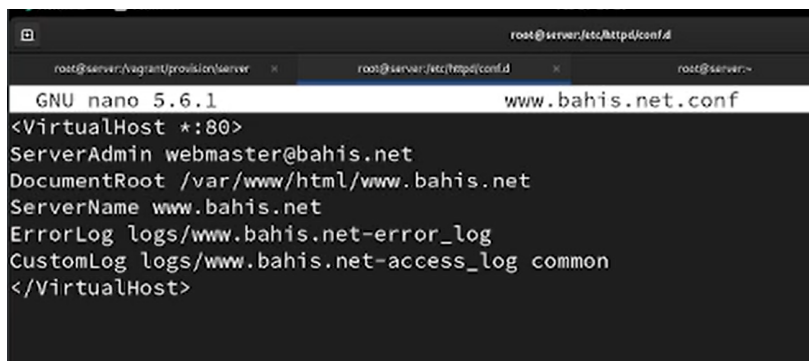


```
GNU nano 5.6.1 server.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/server.bahis.net
ServerName server.bahis.net
ErrorLog logs/server.bahis.net-error_log
CustomLog logs/server.bahis.net-access_log common
</VirtualHost>

I
```

Рисунок 13: Конфигурация виртуального хоста server.bahis.net

- В `www.bahis.net.conf` настроен второй виртуальный хост

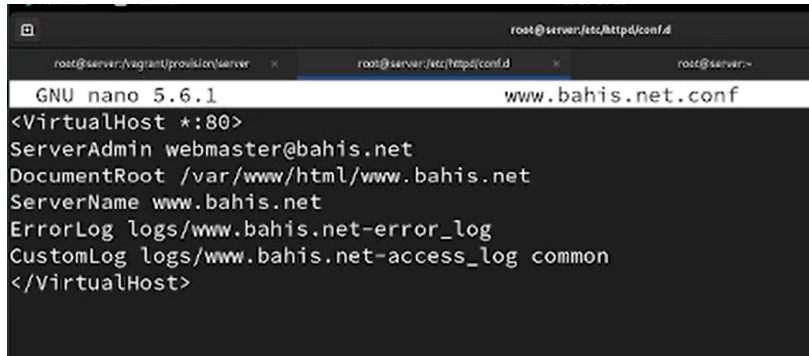


The screenshot shows a terminal window with the nano text editor open. The title bar indicates the file being edited is `root@server:/etc/httpd/conf.d/www.bahis.net.conf`. The editor shows the configuration for a VirtualHost on port 80. The configuration includes the server administrator's email, the document root, the server name, and the paths for error and access logs.

```
GNU nano 5.6.1                               www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
</VirtualHost>
```

Рисунок 14: Конфигурация виртуального хоста `www.bahis.net`

- В `www.bahis.net.conf` настроен второй виртуальный хост
- Отдельный DocumentRoot

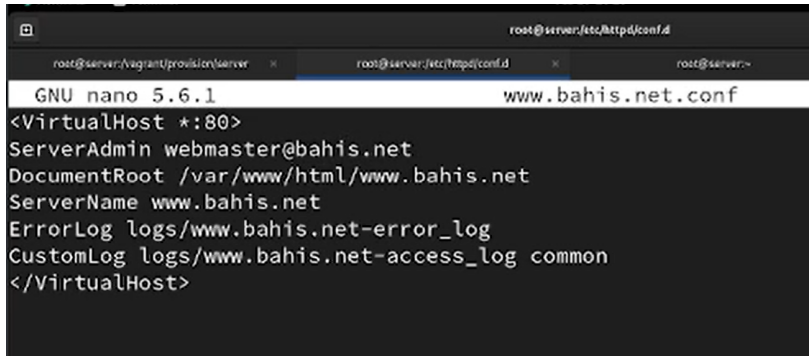


The screenshot shows a terminal window with the nano text editor open. The title bar indicates the file being edited is `root@server:/etc/httpd/conf.d/www.bahis.net.conf`. The editor shows the configuration for a VirtualHost on port 80. The configuration includes the server administrator's email, the document root, the server name, and the paths for error and access logs.

```
GNU nano 5.6.1                               www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
</VirtualHost>
```

Рисунок 14: Конфигурация виртуального хоста `www.bahis.net`

- В `www.bahis.net.conf` настроен второй виртуальный хост
- Отдельный DocumentRoot
- Индивидуальные журналы



The screenshot shows a terminal window with the nano text editor open. The title bar indicates the file being edited is `root@server:/etc/httpd/conf.d/www.bahis.net.conf`. The editor shows the configuration for a VirtualHost on port 80. The configuration includes a ServerAdmin, DocumentRoot, ServerName, and two log files. The terminal window has three tabs open, with the current one being the configuration file.

```
GNU nano 5.6.1 www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
</VirtualHost>
```

Рисунок 14: Конфигурация виртуального хоста `www.bahis.net`

2.4 Создание контента виртуальных хостов

- Создан каталог `/var/www/html/server.bahis.net`

```
[root@server.bahis.net html]# cd /var/www/html
[root@server.bahis.net html]# mkdir server.bahis.net
[root@server.bahis.net html]# cd /var/www/html/server.bahis.net
[root@server.bahis.net server.bahis.net]# touch index.html
[root@server.bahis.net server.bahis.net]# nano index.html
[root@server.bahis.net server.bahis.net]#
```

Рисунок 15: Создание каталога `server.bahis.net` и файла `index.html`

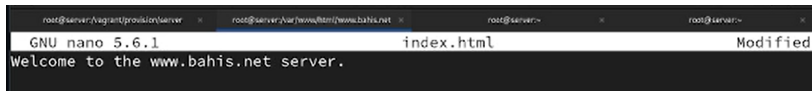
2.4 Создание контента виртуальных хостов

- Создан каталог `/var/www/html/server.bahis.net`
- Создан файл `index.html`

```
[root@server.bahis.net html]# cd /var/www/html
[root@server.bahis.net html]# mkdir server.bahis.net
[root@server.bahis.net html]# cd /var/www/html/server.bahis.net
[root@server.bahis.net server.bahis.net]# touch index.html
[root@server.bahis.net server.bahis.net]# nano index.html
[root@server.bahis.net server.bahis.net]#
```

Рисунок 15: Создание каталога `server.bahis.net` и файла `index.html`

- В `index.html` добавлено тестовое содержимое



```
root@server:/vagrant/provision/server  x  root@server:/var/www/html/www.bahis.net  x  root@server:~  x  root@server:~  x
GNU nano 5.6.1  index.html  Modified
Welcome to the www.bahis.net server.
```

Рисунок 16: Содержимое `index.html` для `server.bahis.net`

- Создан каталог `/var/www/html/www.bahis.net`

```
[root@server.bahis.net html]# cd /var/www/html
[root@server.bahis.net html]# mkdir server.bahis.net
[root@server.bahis.net html]# cd /var/www/html/server.bahis.net
[root@server.bahis.net server.bahis.net]# touch index.html
[root@server.bahis.net server.bahis.net]# nano index.html
[root@server.bahis.net server.bahis.net]#
```

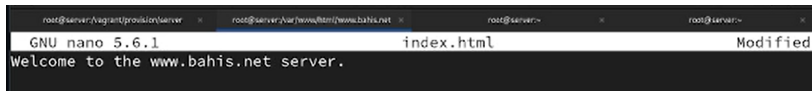
Рисунок 17: Создание каталога `www.bahis.net` и файла `index.html`

- Создан каталог `/var/www/html/www.bahis.net`
- Создан файл `index.html`

```
[root@server.bahis.net html]# cd /var/www/html
[root@server.bahis.net html]# mkdir server.bahis.net
[root@server.bahis.net html]# cd /var/www/html/server.bahis.net
[root@server.bahis.net server.bahis.net]# touch index.html
[root@server.bahis.net server.bahis.net]# nano index.html
```

Рисунок 17: Создание каталога `www.bahis.net` и файла `index.html`

- В `index.html` добавлено тестовое содержимое



```
root@server:/vagrant/provision/server  x  root@server:/var/www/html/www.bahis.net  x  root@server:~  x  root@server:~  x
GNU nano 5.6.1  index.html  Modified
Welcome to the www.bahis.net server.
```

Рисунок 18: Содержимое `index.html` для `www.bahis.net`

2.5 Настройка прав доступа и SELinux

- Выполнен `chown -R apache:apache /var/www`

```
[root@server.bahis.net www.bahis.net]# chown -R apache:apache /var/www
[root@server.bahis.net www.bahis.net]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
[root@server.bahis.net www.bahis.net]# restorecon -vR /var/named
[root@server.bahis.net www.bahis.net]# restorecon -vR /var/www
[root@server.bahis.net www.bahis.net]# systemctl restart httpd
[root@server.bahis.net www.bahis.net]#
```

Рисунок 19: Корректировка прав доступа, восстановление контекста SELinux и перезапуск httpd

2.5 Настройка прав доступа и SELinux

- Выполнен `chown -R apache:apache /var/www`
- Восстановлены SELinux-контексты

```
[root@server.bahis.net www.bahis.net]# chown -R apache:apache /var/www
[root@server.bahis.net www.bahis.net]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:
net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_
t:s0 to unconfined_u:object_r:net_conf_t:s0
[root@server.bahis.net www.bahis.net]# restorecon -vR /var/named
[root@server.bahis.net www.bahis.net]# restorecon -vR /var/www
[root@server.bahis.net www.bahis.net]# systemctl restart httpd
[root@server.bahis.net www.bahis.net]#
```

Рисунок 19: Корректировка прав доступа, восстановление контекста SELinux и перезапуск httpd

2.5 Настройка прав доступа и SELinux

- Выполнен `chown -R apache:apache /var/www`
- Восстановлены SELinux-контексты
- Перезапущен `httpd`

```
[root@server.bahis.net www.bahis.net]# chown -R apache:apache /var/www
[root@server.bahis.net www.bahis.net]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
[root@server.bahis.net www.bahis.net]# restorecon -vR /var/named
[root@server.bahis.net www.bahis.net]# restorecon -vR /var/www
[root@server.bahis.net www.bahis.net]# systemctl restart httpd
[root@server.bahis.net www.bahis.net]#
```

Рисунок 19: Корректировка прав доступа, восстановление контекста SELinux и перезапуск `httpd`

2.6 Проверка работы виртуальных хостов

- На клиенте открыт `http://server.bahis.net`



Рисунок 20: Доступ к виртуальному хосту server.bahis.net

2.6 Проверка работы виртуальных хостов

- На клиенте открыт `http://server.bahis.net`
- Отобрана тестовая страница



Рисунок 20: Доступ к виртуальному хосту server.bahis.net

- Открыт `http://www.bahis.net`



Рисунок 21: Доступ к виртуальному хосту `www.bahis.net`

- Открыт `http://www.bahis.net`
- Отобрана вторая тестовая страница



Рисунок 21: Доступ к виртуальному хосту `www.bahis.net`

2.7 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создана структура `http/etc/httpd/conf.d`

```
[root@server.bahis.net www.bahis.net]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server.bahis.net server]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http
/etc/httpd/conf.d/
[root@server.bahis.net server]# cp -R /var/www/html/* /vagrant/provision/server/http/var
/www/html
[root@server.bahis.net server]# cd /vagrant/provision/server/dns/
[root@server.bahis.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net dns]# cd /vagrant/provision/server
025:38 [root@server.bahis.net server]# touch http.sh
[root@server.bahis.net server]# chmod +x http.sh
```

2.7 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создана структура `http/etc/httpd/conf.d`
- Создана структура `http/var/www/html`

```
[root@server.bahis.net www.bahis.net]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server.bahis.net server]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http
/etc/httpd/conf.d/
[root@server.bahis.net server]# cp -R /var/www/html/* /vagrant/provision/server/http/var
/www/html
[root@server.bahis.net server]# cd /vagrant/provision/server/dns/
[root@server.bahis.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net dns]# cd /vagrant/provision/server
025:38 [root@server.bahis.net server]# touch http.sh
[root@server.bahis.net server]# chmod +x http.sh
```

2.7 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создана структура `http/etc/httpd/conf.d`
- Создана структура `http/var/www/html`
- Скопированы конфигурации и контент

```
[root@server.bahis.net www.bahis.net]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server.bahis.net server]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http
/etc/httpd/conf.d/
[root@server.bahis.net server]# cp -R /var/www/html/* /vagrant/provision/server/http/var
/www/html
[root@server.bahis.net server]# cd /vagrant/provision/server/dns/
[root@server.bahis.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net dns]# cd /vagrant/provision/server
025:38 [root@server.bahis.net server]# touch http.sh
[root@server.bahis.net server]# chmod +x http.sh
```

- Скопированы файлы DNS-зон в provision

```
[root@server.bahis.net www.bahis.net]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server.bahis.net server]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http
/etc/httpd/conf.d/
[root@server.bahis.net server]# cp -R /var/www/html/* /vagrant/provision/server/http/var
/www/html
[root@server.bahis.net server]# cd /vagrant/provision/server/dns/
[root@server.bahis.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net dns]# cd /vagrant/provision/server
[0:25:38 root@server.bahis.net server]# touch http.sh
[root@server.bahis.net server]# chmod +x http.sh
[root@server.bahis.net server]# nano http.sh
```

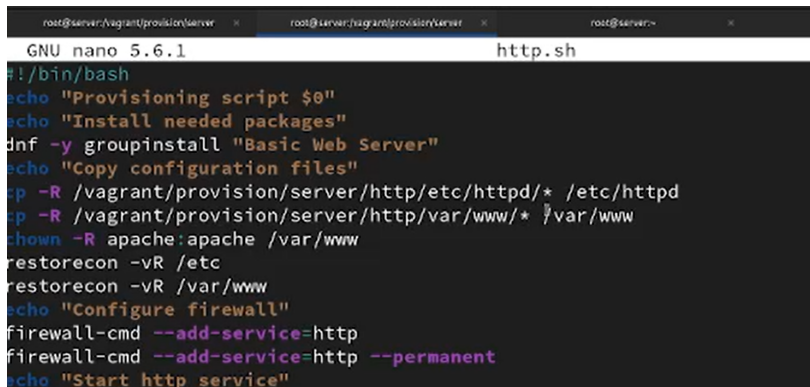
Рисунок 23: Копирование файлов DNS-зон в каталог provision

- Создан исполняемый файл http.sh

```
[root@server.bahis.net www.bahis.net]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server.bahis.net server]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http
/etc/httpd/conf.d/
[root@server.bahis.net server]# cp -R /var/www/html/* /vagrant/provision/server/http/var
/www/html
[root@server.bahis.net server]# cd /vagrant/provision/server/dns/
[root@server.bahis.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.bahis.net dns]# cd /vagrant/provision/server
[roo0:25:38@server.bahis.net server]# touch http.sh
[root@server.bahis.net server]# chmod +x http.sh
[root@server.bahis.net server]# nano http.sh
```

Рисунок 24: Создание и подготовка файла http.sh

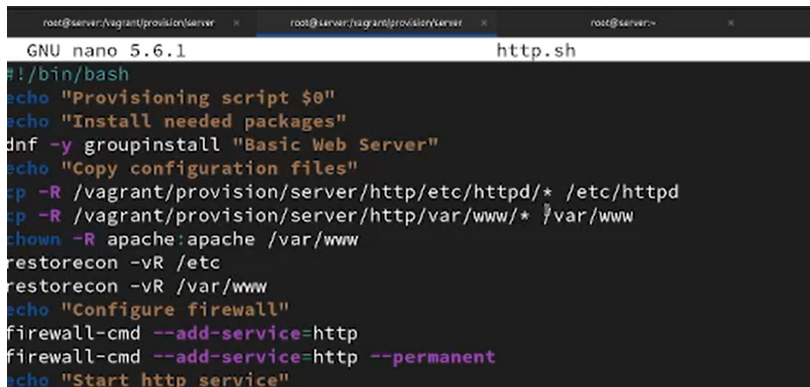
- В `http.sh` реализованы:



The screenshot shows a terminal window with three tabs. The active tab is titled `root@server:~`. The terminal is running the `GNU nano 5.6.1` editor, editing a file named `http.sh`. The script content is as follows:

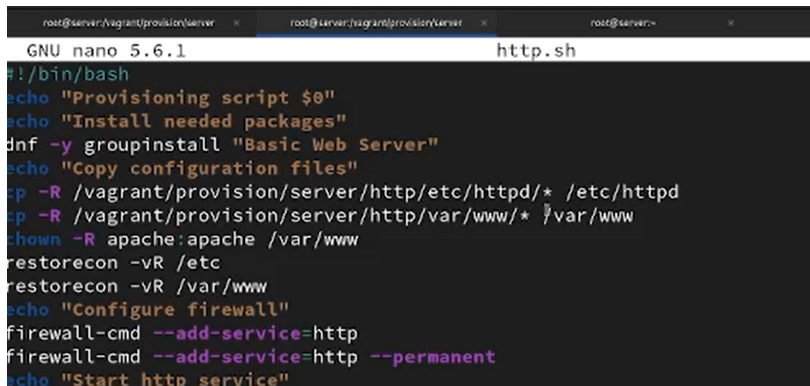
```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```

- В `http.sh` реализованы:
- установка Basic Web Server



```
root@server:/vagrant/provision/server x root@server:/vagrant/provision/server x root@server:~ x
GNU nano 5.6.1 http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```

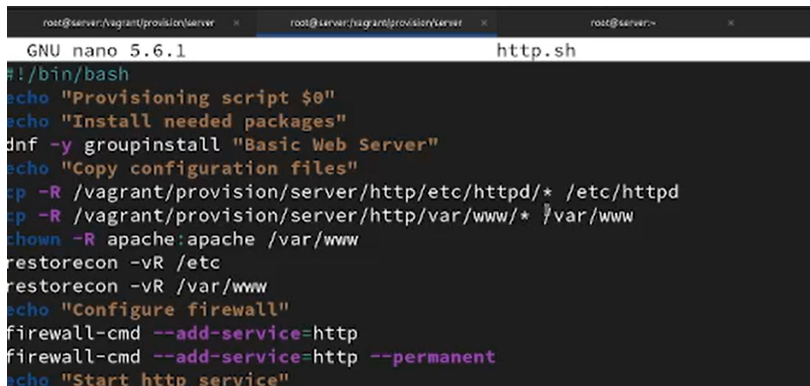
- В `http.sh` реализованы:
- установка Basic Web Server
- копирование конфигураций



The screenshot shows a terminal window with three tabs. The active tab is titled `root@server:/vagrant/provision/server`. The terminal is running GNU nano 5.6.1, editing a file named `http.sh`. The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```

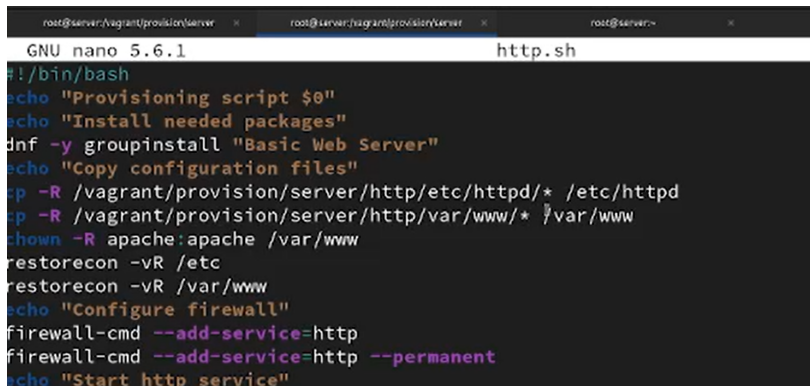
- В `http.sh` реализованы:
- установка Basic Web Server
- копирование конфигураций
- назначение владельца `apache`



The screenshot shows a terminal window with three tabs. The active tab is titled `root@server:/vagrant/provision/server`. The terminal displays the content of a file named `http.sh`, which is being edited with GNU nano 5.6.1. The script contains the following commands:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```

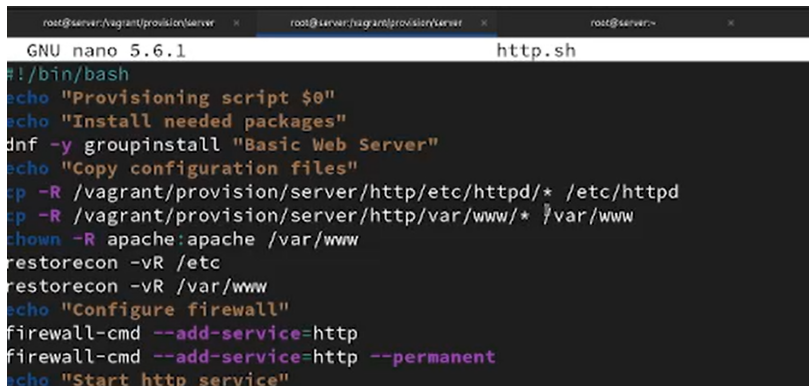
- В `http.sh` реализованы:
- установка Basic Web Server
- копирование конфигураций
- назначение владельца `apache`
- восстановление SELinux



The screenshot shows a terminal window with three tabs. The active tab is titled `root@server:/vagrant/provision/server`. The terminal displays the content of a file named `http.sh`, which is being edited with GNU nano 5.6.1. The script contains the following commands:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```

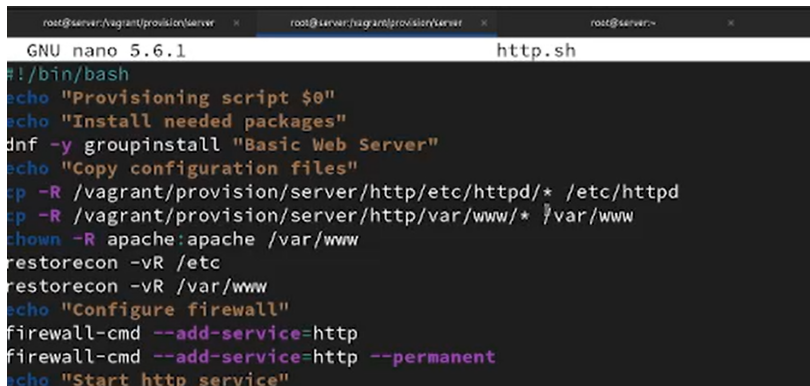
- В `http.sh` реализованы:
- установка Basic Web Server
- копирование конфигураций
- назначение владельца `apache`
- восстановление SELinux
- настройка `firewalld`



The screenshot shows a terminal window with three tabs. The active tab is titled `root@server:/vagrant/provision/server`. The terminal displays the content of a file named `http.sh`, which is being edited with GNU nano 5.6.1. The script contains the following commands:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```

- В `http.sh` реализованы:
- установка Basic Web Server
- копирование конфигураций
- назначение владельца `apache`
- восстановление SELinux
- настройка `firewalld`
- запуск `httpd`



The screenshot shows a terminal window with three tabs. The active tab is titled `root@server:/vagrant/provision/server`. The terminal content is as follows:

```
GNU nano 5.6.1 http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
```


- В Vagrantfile добавлен provisioner shell

```
server.vm.provision "server http",  
type: "shell",  
preserve_order: true,  
path: "provision/server/http.sh"
```

Рисунок 26: Добавление provisioner в Vagrantfile

- В Vagrantfile добавлен provisioner shell
- Указан путь provision/server/http.sh

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- В Vagrantfile добавлен provisioner shell
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- Установлен preserve_order: true

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Рисунок 26: Добавление provisioner в Vagrantfile

Раздел 3

3. Выводы

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- Проверена доступность сайтов с клиента
- Реализована автоматизация через `http.sh` и `Vagrantfile`